Republic of the Philippines
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
COLLEGE OF ENGINEERING
COMPUTER ENGINEERING DEPARTMENT

BSCOE-ELEC2

(DATABASE MANAGEMENT SYSTEMS)

EXERCISE 3

USING DDL STATEMENTS TO CREATE AND MANAGE TABLES

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BSCPE 4-7
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I. OBJECTIVES:

At the end of this exercise, students must be able to:

a) Create, alter and drop tables
b) Verify that tables exist
c) Set a table to read-only and read/write status
d) Manage constraints
e) Add, modify and drop columns
f) Perform flashback operations

II. BACKGROUND INFORMATION

CREATE TABLE Statement

- This statement is one of the DDL statements that are a subset of the SQL statements used to create, modify, or remove Oracle database structures.

Syntax:

```
CREATE TABLE [schema.]table
    (column datatype [DEFAULT expr] [, …]);
```

Constraints

The Oracle server uses constraints to prevent invalid data entry into tables.

You can use constraints to do the following:

- Enforce rules on the data in a table whenever a row is inserted, updated, or deleted from that table. The constraint must be satisfied for the operation to succeed.
- Prevent the deletion of a table if there are dependencies from other tables.
- Provide rules for Oracle tools, such as Oracle Developer.

Data Integrity Constraints

Constraint	Description
NOT NULL	Specifies that the column cannot contain a null value
UNIQUE	Specifies a column or combination of columns whose values must be unique for all rows in the table
PRIMARY KEY	Uniquely identifies each row of the table
FOREIGN KEY	Establishes and enforces a referential integrity between the column and a column of the referenced table such that values in one table match values in another table.
CHECK	Specifies a condition that must be true

Defining Constraints

Syntax:

```
CREATE TABLE [schema.]table
    (column datatype [DEFAULT expr]
    [column_constraint],
    …
    [table_constraint] [, …]);
```

- Column-level constraint syntax:

  Syntax:

  ```
  column [CONSTRAINT constraint_name] constraint_type,
  ```

- Table-level constraint syntax:

  Syntax:

  ```
  column, …
      [CONSTRAINT constraint_name] constraint_type
      (column, …),
  ```

Creating a Table Using a Subquery

- Match the number of specified columns to the number of subquery columns.
- Define columns with column names and default values.

Syntax:

```
CREATE TABLE table
    [(column, column…)]
AS subquery;
```

Read-Only Tables

You can use the ALTER TABLE syntax to:

- Put a table into read-only mode, which prevents DDL or DML changes during table maintenance
- Put the table back into read/write mode

Syntax:

```
ALTER TABLE table READ ONLY;
ALTER TABLE table READ WRITE;
```

Dropping a Table

- Moves a table to the recycle bin
- Removes the table and all its data entirely if the PURGE clause is specified

Syntax:

DROP TABLE *table* [PURGE];

Inserting Data into the Table

- With this syntax, only one row is inserted at a time

Syntax:

INSERT INTO *table* [(*column* [, *column*…])]
VALUES (*value* [, *value*…]);

- Copying rows from another table

Syntax:

INSERT INTO *table* [(*column* [, *column*…])]
subquery;

ALTER TABLE Statement

Syntax:

ALTER TABLE *table*
ADD (*column datatype* [DEFAULT *expr*]
[, *column datatype*]…);

ALTER TABLE *table*
MODIFY (*column datatype* [DEFAULT *expr*]
[, *column datatype*]…);

ALTER TABLE *table*
DROP (*column*);

Adding a Constraint

Syntax:

ALTER TABLE *table*
ADD (CONSTRAINT *constraint_name*] *type column_name*;

Dropping a Constraint

Syntax:

ALTER TABLE *table*
DROP PRIMARY KEY | UNIQUE (*column*) |
CONSTRAINT *constraint* [CASCADE];

FLASHBACK TABLE Statement

III. EXPERIMENTAL PROCEDURE

Overview

In this exercise, you are to create new tables by using the CREATE TABLE statement, ALTER TABLE command to modify columns and add constraints, and confirm that the new table was added to the database. You will also set the status of a table as READ ONLY and then revert to READ/WRITE.

Note: You will use the DEPARTMENTS and EMPLOYEES table in the HR schema. Use a qualifier to query data from the table in the HR schema.

Syntax: <schema>.table Example: HR.DEPARTMENTS

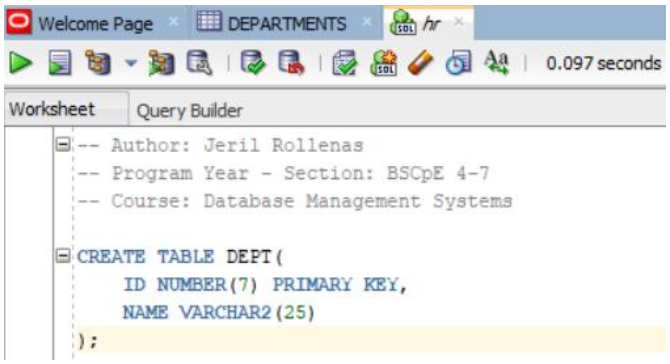
Task

Write the equivalent SQL statements for the steps that follow.

Step 1: Create a DEPT table based on the following table instance chart. Enter the syntax in the SQL Worksheet. Then, execute the statement to create the table.

Column Name	ID	NAME
Data Type	NUMBER	VARCHAR
Length	7	25

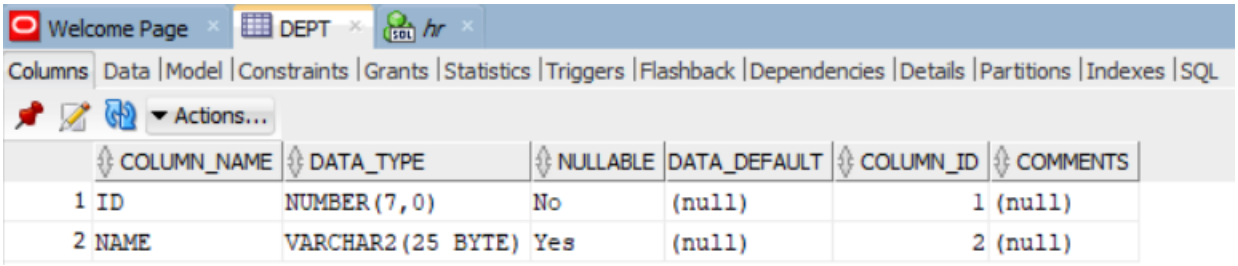
Query used:



Step 2: Confirm that the table is created. Sample output is as follows:

```
Name                          Null     Type
----------------------------- -------- ----------------
ID                            NOT NULL NUMBER(7)
NAME                                   VARCHAR2(25)
```

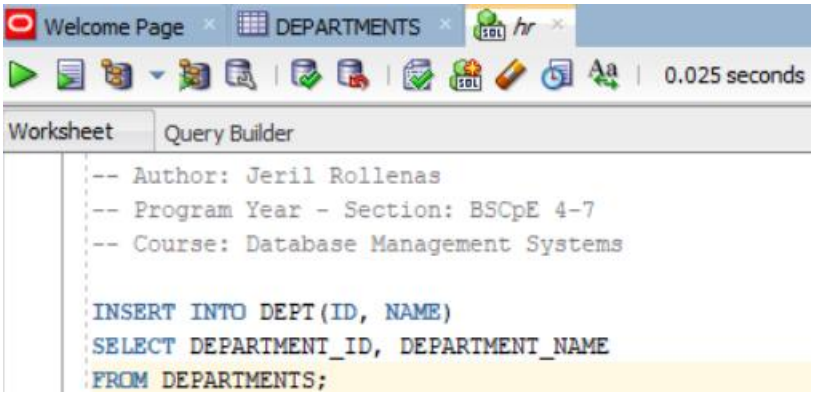
Script Output for Step 2:



	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID	NUMBER(7,0)	No	(null)	1	(null)
2	NAME	VARCHAR2(25 BYTE)	Yes	(null)	2	(null)

Step 3: Populate the DEPT table with data from the DEPARTMENTS table. Include only columns that you need.

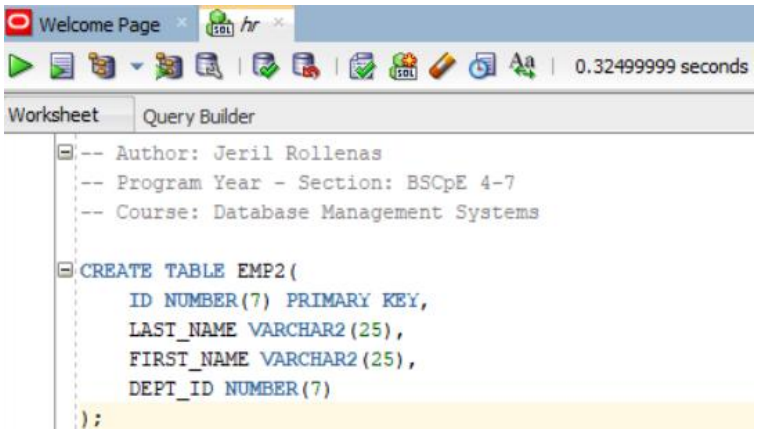
Query used:



Step 4: Create the EMP2 table based on the following table instance chart. Enter the syntax in the SQL Worksheet. Then, execute the statement to create the table.

Column name	ID	LAST_NAME	FIRST_NAME	DEPT_ID
Data type	NUMBER	VARCHAR2	VARCHAR2	NUMBER
Length	7	25	25	7

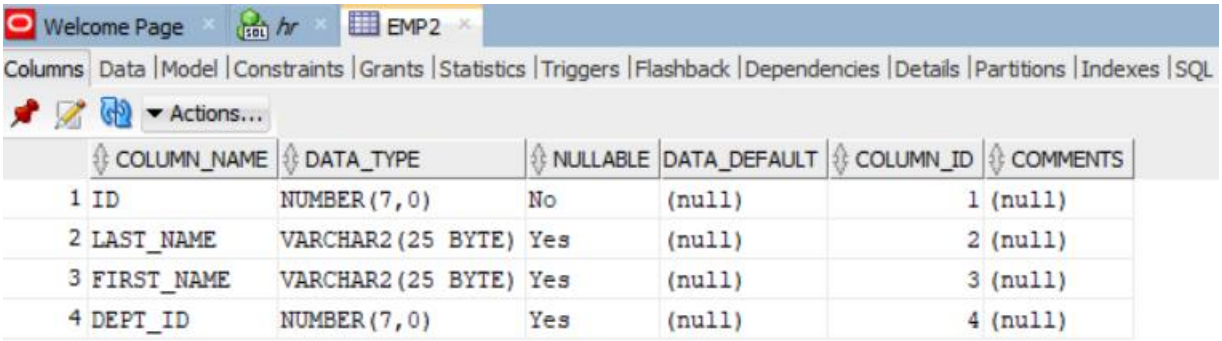
Query Used:



Step 5: Confirm that the table is created. Sample output is as follows.

```
Name                          Null     Type
----------------------------- -------- ------------
ID                                     NUMBER(7)
LAST_NAME                              VARCHAR2(25)
FIRST_NAME                             VARCHAR2(25)
DEPT_ID                                NUMBER(7)
```

Script Output for Step 5:



	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID	NUMBER(7,0)	No	(null)	1	(null)
2	LAST_NAME	VARCHAR2(25 BYTE)	Yes	(null)	2	(null)
3	FIRST_NAME	VARCHAR2(25 BYTE)	Yes	(null)	3	(null)
4	DEPT_ID	NUMBER(7,0)	Yes	(null)	4	(null)

Step 6: Modify the EMP2 table to allow for longer employee last names.

Query Used :

```
-- Author: Jeril Rollenas
-- Program Year - Section: BSCpE 4-7
-- Course: Database Management Systems

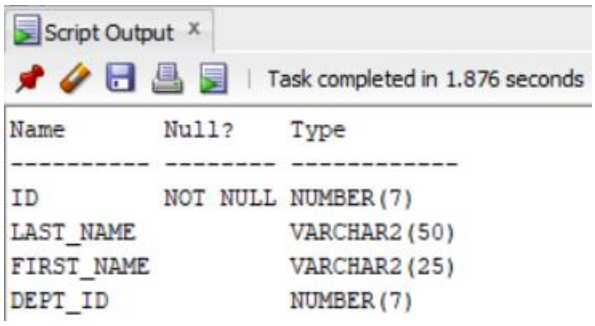
ALTER TABLE EMP2
MODIFY(LAST_NAME VARCHAR(50));
```

Step 7: Confirm your modification. Sample output is as follows.

```
Name                          Null     Type
----------------------------- -------- ------------------
ID                                     NUMBER(7)
LAST_NAME                              VARCHAR2(50)
FIRST_NAME                             VARCHAR2(25)
DEPT_ID                                NUMBER(7)

4 rows selected
```

Script Output:



Step 8: Create the EMPLOYEES2 table based on the structure of the EMPLOYEES table. Include only the EMPLOYEE_ID, FIRST_NAME, LAST_NAME, SALARY, and DEPARTMENT_ID columns. Name the columns in your new table ID, FIRST_NAME, LAST_NAME, SALARY, and DEPT_ID, respectively.

Query Used:

```
-- Author: Jeril Rollenas
-- Program Year - Section: BSCpE 4-7
-- Course: Database Management Systems

CREATE TABLE EMPLOYEES2(
    ID NUMBER(6),
    FIRST_NAME VARCHAR(20),
    LAST_NAME VARCHAR(25),
    SALARY NUMBER(8,2),
    DEPT_ID number(4)
);
```

Step 9: Alter the EMPLOYEES2 table status to read-only.

Query Used:

```
ALTER TABLE EMPLOYEES2 READ ONLY;
```

Step 10: Try to insert the following row into the EMPLOYEES2 table:

ID	FIRST_NAME	LAST_NAME	SALARY	DEPT_ID
34	Grant	Marcie	5678	10

You get the following error message:

```
Error starting at line 1 in command:
INSERT INTO employees2
VALUES (34, 'Grant','Marcie',5678,10)
Error at Command Line:1 Column:12
Error report:
SQL Error: ORA-12081: update operation not allowed on table "ORA16"."EMPLOYEES2"
12081. 00000 -  "update operation not allowed on table \"%s\".\"%s\""
*Cause:    An attempt was made to update a read-only materialized view.
*Action:   No action required. Only Oracle is allowed to update a
           read-only materialized view.
```

Query Used:

```
-- Author: Jeril Rollenas
-- Program Year - Section: BSCpE 4-7
-- Course: Database Management Systems

INSERT INTO EMPLOYEES2(ID, FIRST_NAME, LAST_NAME, SALARY, DEPT_ID)
VALUES(34, 'GRANT', 'MARCIE', 5678, 10);
```

Script Output:

```
Error starting at line : 5 in command -
INSERT INTO EMPLOYEES2(ID, FIRST_NAME, LAST_NAME, SALARY, DEPT_ID)
VALUES(34, 'GRANT', 'MARCIE', 5678, 10)
Error at Command Line : 5 Column : 13
Error report -
SQL Error: ORA-12081: update operation not allowed on table "HR"."EMPLOYEES2"

https://docs.oracle.com/error-help/db/ora-12081/12081. 00000 -  "update operation not allowed on table \"%s\".\"%s\""
*Cause:    An attempt was made to update a read-only materialized view.
*Action:   No action required. Only Oracle is allowed to update a
           read-only materialized view.

More Details :
https://docs.oracle.com/error-help/db/ora-12081/
```

Step 11: Revert the EMPLOYEES2 table to the read/write status. Now, try to insert the same row again. You should get the following messages:

```
ALTER TABLE employees2 succeeded.
1 rows inserted
```

Query Used:

```
-- Author: Jeril Rollenas
-- Program Year - Section: BSCpE 4-7
-- Course: Database Management Systems

ALTER TABLE EMPLOYEES2 READ WRITE;
INSERT INTO EMPLOYEES2(ID, FIRST_NAME, LAST_NAME, SALARY, DEPT_ID)
VALUES(34, 'GRANT', 'MARCIE', 5678, 10);
```

Script Output:

```
Table EMPLOYEES2 altered.


1 row inserted.
```

Step 12: Drop the EMP2 table.

Query Used:

```
DROP TABLE EMP2;
```

Step 13: Query the recycle bin to see whether the table is present. Sample output is as follows.

	ORIGINAL_NAME	OPERATION	DROPTIME
1	EMP2	DROP	2007-07-02:04:52:43

Query Used:

```
-- Author: Jeril Rollenas
-- Program Year - Section: BSCpE 4-7
-- Course: Database Management Systems

SELECT ORIGINAL_NAME, OPERATION, DROPTIME FROM USER_RECYCLEBIN WHERE TYPE = 'TABLE';
```

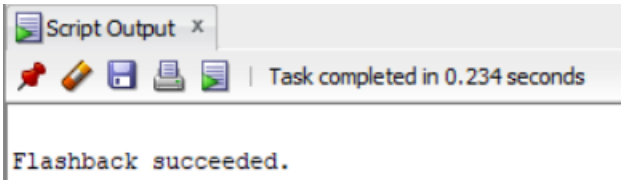
Script Output:

```
Script Output
Task completed in 0.06 seconds

ORIGINAL_NAME                                                
```

FLASHBACK TABLE EMP2 TO BEFORE DROP;

Script Output:



Step 15: Drop the first name column from the EMPLOYEES2 table.

Query Used:

ALTER TABLE EMPLOYEES2 DROP(FIRSTNAME);

Step 16: Confirm your modification by checking the description of the table. Sample output is as follows.

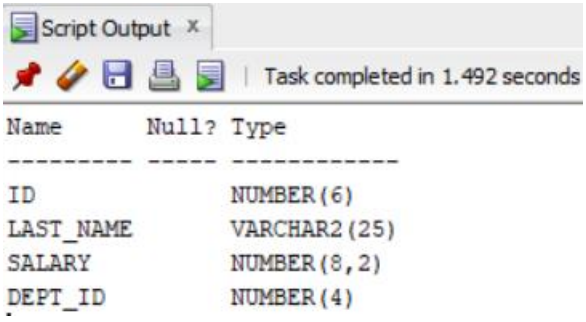
Name	Null	Type
ID		NUMBER(6)
LAST_NAME	NOT NULL	VARCHAR2(25)
SALARY		NUMBER(8,2)
DEPT_ID		NUMBER(4)

4 rows selected

Query Used:

DESC EMPLOYEES2;

Script Output:



Step 17: Add a table-level PRIMARY KEY constraint on the EMP table on the ID column. The constraint should be named at creation. Name the constraint my_emp_id_pk.

Query Used:

ALTER TABLE EMP2 ADD CONSTRAINT my_emp_id_pk PRIMARY KEY (ID);

Step 18: Create a PRIMARY KEY constraint to the DEPT table using the ID column. The constraint should be named at creation. Name the constraint my_dept_id_pk.

Query Used:

ALTER TABLE EMP2 ADD CONSTRAINT my_emp_id_pk PRIMARY KEY (ID);

Step 19: Add a foreign key reference on the EMP2 table that ensures that the employee is not assigned to a nonexistent department. Name the constraint my_emp_dept_id_fk.

Query Used:

```
ALTER TABLE EMP2 ADD CONSTRAINT my_emp_dept_id_fk FOREIGN KEY (DEPT_ID)
REFERENCES DEPT (ID);
```

Step 20: Modify the EMP2 table. Add a COMMISSION column of the NUMBER data type, precision 2, scale 2. Add a constraint to the COMMISSION column that ensures that a commission value is greater than zero.

Query Used:

```
ALTER TABLE EMP2
ADD(
   COMMISSION NUMBER (2,2)
   CONSTRAINT greaterthan_zero_value
      CHECK(COMMISSION > 0)
);
```

Step 21: Drop the EMP2 and DEPT tables so that they cannot be restored.

Query Used:

```
DROP TABLE EMP2 PURGE;
DROP TABLE DEPT PURGE;
```

Step 22: Verify the recycle bin.

What do you observe? Recycle bin is empty because of PURGE script.

IV. QUESTION AND ANSWER

1. Explain why you encountered an error message in Step 10 after you inserted a row into the EMPLOYEES2 table.
   - EMPLOYEES2 table was set to READ ONLY mode, so data insertion was disabled.

2. What is the purpose of modifying the status of your table to READ ONLY or READ/WRITE mode?
   - READ ONLY protects important reference-only tables by only allowing it to be read.
   - READ/WRITE allows insertion and deletion of data inserted on the tables.